

**Question:**  
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A.  $2.48 \times 10^{-8}$  cm
  - B.  $5.92 \times 10^7$  cm/sec
  - C.  $8.66 \times 10^{-6}$  cm
  - D.  $3.14 \times 10^{-5}$  cm
  - E.  $9.81 \times 10^{-4}$  cm
  - F.  $5.27 \times 10^{-9}$  m
  - G.  $1.23 \times 10^{-7}$  cm
  - H.  $4.14 \times 10^{-10}$  m
  - I.  $6.625 \times 10^{-27}$  erg sec
  - J.  $0.512 \times 10^6$ eV

**Reasoning + Answer:**

<think>  
Hereâ€™s a concise solution outline :

- Kin etic energy :  $E = 1 \text{ eV} \hat{=} 1.602 \times 10^{-12} \text{ erg} .$
- Relate kinetic energy to momentum using  $E = p^2/(2m)$ , solving for p gives  $p = \sqrt{2mE}$ .
- Mass of electron :  $m_e \hat{=} 9.109 \times 10^{-28} \text{ g} .$
- Compute momentum :  $p = \sqrt{2 \times (9.109 \times 10^{-28} \text{ g}) \times (1.602 \times 10^{-12} \text{ erg})} \hat{=} 4.36 \times 10^{-21} \text{ g}\cdot\text{cm/s} .$
- De Broglie wavelength :  $\lambda = h/p .$
- Planckâ€™s constant in the appropriate units for cm :  $h \hat{=} 6.625 \times 10^{-27} \text{ erg}\cdot\text{sec} .$
- Thus  $\lambda \hat{=} (6.625 \times 10^{-27}) / (4.36 \times 10^{-21}) \hat{=} 1.52 \times 10^{-6} \text{ cm} \hat{=} 1.23 \times 10^{-7} \text{ cm} .$

Therefore , the correct answer is : G .  $1.23 \times 10^{-7} \text{ cm} .$

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<answer>  
G .  $1.23 \times 10^{-7} \text{ cm}$   
</answer>